

Annual Infrastructure Compliance Report

Document Classification & OCR Performance Analysis

Report Date: 2025-06-15 | Classification: INTERNAL | Version: 3.2

Executive Summary

This report summarizes the infrastructure compliance status across all Azure regions. Key metrics include resource utilization rates, security posture scores, cost optimization opportunities, and SLA adherence. The OCR extraction pipeline processed 12,847 documents with a 99.2% accuracy rate on structured content and 97.8% on handwritten annotations.

The Mistral Document AI model was deployed on Azure AI Foundry in West Europe, processing an average of 1,285 documents per day. Table extraction accuracy reached 99.1% on printed content. The multi-stage classification pipeline achieves an overall F1-score of 0.983 across 10 document types.

Table 1: Key Performance Indicators

KPI	Target	Actual	Status	Trend
OCR Accuracy (structured)	99.0%	99.2%	PASS	Improving
OCR Accuracy (handwritten)	95.0%	97.8%	PASS	Stable
Processing Time p95	<5s	3.2s	PASS	Improving
Documents Processed/Day	1,000	1,285	PASS	Increasing
Error Rate	<1.0%	0.3%	PASS	Decreasing
API Availability	99.9%	99.97%	PASS	Stable
Cost per Document	<\$0.05	\$0.032	PASS	Decreasing
Model Drift Score	<0.15	0.08	PASS	Stable

Table 2: Regional Deployment Status

Region	Status	Instances	CPU %	Memory %	Cost/mo
West Europe	Active	4	62%	71%	\$2,340
North Europe	Active	2	45%	58%	\$1,120
East US	Active	3	78%	82%	\$1,950
Southeast Asia	Standby	1	12%	25%	\$380
Australia East	Active	2	55%	63%	\$1,080
UK South	Migrating	2	31%	44%	\$920

Section 2: Cost Analysis

Table 3: Detailed Cost Breakdown by Service (Monthly)

Service	SKU	Qty	Unit Price	Subtotal	Discount	Net Cost	vs Budget
Container Apps	Consumption	4	\$0.000012/s	\$1,244.80	10%	\$1,120.32	-5.2%
Cosmos DB	Serverless	1	\$0.25/RU	\$892.50	0%	\$892.50	+2.1%
Service Bus	Standard	3	\$0.05/msg	\$456.30	15%	\$387.86	-12.4%
Blob Storage	Hot	500GB	\$0.018/GB	\$9.00	0%	\$9.00	-8.3%
Key Vault	Standard	2	\$0.03/op	\$45.60	0%	\$45.60	+1.2%
AI Foundry	S0	1	\$1.50/1k	\$3,450.00	20%	\$2,760.00	-3.8%
App Insights	Pay-as-go	1	\$2.30/GB	\$234.60	0%	\$234.60	+5.7%
Container Reg	Basic	1	\$5.00/day	\$150.00	0%	\$150.00	0.0%
DNS Zone	Public	2	\$0.50/zone	\$1.00	0%	\$1.00	0.0%
Load Balancer	Standard	1	\$0.025/hr	\$18.00	0%	\$18.00	-2.1%
Virtual Network	Basic	2	\$0.00	\$0.00	0%	\$0.00	0.0%
Managed Identity	-	3	\$0.00	\$0.00	0%	\$0.00	0.0%
Log Analytics	Pay-as-go	1	\$2.76/GB	\$345.00	5%	\$327.75	+4.3%
TOTAL				\$6,846.80		\$5,946.63	-2.7%

Table 4: SLA Compliance Matrix

Service	SLA Target	Jan	Feb	Mar	Apr	May	Jun
API Gateway	99.95%	99.98%	99.97%	99.99%	99.96%	99.98%	99.99%
OCR Pipeline	99.90%	99.95%	99.92%	99.94%	99.91%	99.93%	99.97%
Database	99.99%	100.0%	99.99%	100.0%	100.0%	99.99%	100.0%
Queue System	99.95%	99.99%	99.98%	99.97%	99.99%	99.96%	99.99%
Storage	99.99%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Auth Service	99.99%	99.99%	100.0%	99.99%	100.0%	99.99%	100.0%

Notes:

- 1. All percentages represent uptime measured on 5-minute intervals.
- 2. Discount rates are negotiated enterprise agreement terms.
- 3. Budget variance is calculated against Q2 2025 forecast.
- 4. Cost optimization recommendations are in Appendix B.

Section 3: Document Classification Results

The classification pipeline categorizes incoming documents into predefined types using a multi-stage approach: (1) OCR extraction via Mistral Document AI, (2) structural analysis of headers, tables, and layout, (3) LLM-based semantic classification using GPT-4.1-mini, and (4) confidence scoring with human review for low-confidence results.

Table 5: Classification Accuracy by Document Type

Document Type	Total	Correct	Accuracy	Precision	Recall	F1-Score
Invoice	2,345	2,312	98.6%	0.987	0.986	0.986
Purchase Order	1,890	1,852	97.9%	0.981	0.979	0.980
Delivery Note	1,456	1,428	98.1%	0.983	0.981	0.982
Contract	987	968	98.1%	0.984	0.978	0.981
Insurance Claim	654	639	97.7%	0.979	0.977	0.978
Tax Document	1,234	1,216	98.5%	0.987	0.985	0.986
Bank Statement	876	862	98.4%	0.986	0.984	0.985
Medical Record	543	527	97.1%	0.974	0.971	0.972
Legal Filing	432	420	97.2%	0.975	0.972	0.973
Correspondence	2,430	2,388	98.3%	0.985	0.983	0.984

Table 6: Error Analysis - Top Misclassifications

Actual Type	Predicted As	Count	% of Errors	Root Cause
Purchase Order	Invoice	18	12.3%	Similar header layout
Delivery Note	Invoice	14	9.6%	Shared vendor template
Contract	Legal Filing	11	7.5%	Legal terminology overlap
Insurance Claim	Medical Record	9	6.2%	Medical content in claims
Correspondence	Contract	8	5.5%	Formal letter format
Bank Statement	Invoice	7	4.8%	Tabular financial data
Tax Document	Invoice	6	4.1%	Numerical data patterns

Section 4: Infrastructure Configuration

Table 7: Container App Environment Variables

Variable	Service	Source	Description
COSMOS_ENDPOINT	API + Worker	Terraform	Cosmos DB account endpoint
COSMOS_DATABASE	API + Worker	Terraform	Database name (classymail)
COSMOS_CONTAINER	API + Worker	Terraform	Container name (emails)
SERVICEBUS_FQDN	API + Worker	Terraform	Service Bus namespace FQDN
SERVICEBUS_QUEUE	Worker	Terraform	Queue name (pdf-queue)
STORAGE_ACCOUNT	API + Worker	Terraform	Blob storage account name
STORAGE_CONTAINER	API + Worker	Terraform	Blob container (attachments)
AI_FOUNDRY_ENDPOINT	API + Worker	Terraform	AI Foundry project endpoint
AI_FOUNDRY_KEY	API + Worker	Key Vault	API key (managed identity)
MISTRAL_DEPLOYMENT	Worker	Terraform	OCR model deployment name
GPT_DEPLOYMENT	API + Worker	Terraform	LLM deployment name
APPLICATIONINSIGHTS_CS	API + Worker	Terraform	App Insights conn string
AZURE_CLIENT_ID	API + Worker	Terraform	UAMI client ID for RBAC
ASPNETCORE_URLS	API	Static	Listen port configuration

Table 8: RBAC Role Assignments

Principal	Role	Scope	Purpose
UAMI (api)	Cosmos DB Data Contrib	Cosmos Account	Read/write docs
UAMI (api)	Storage Blob Reader	Storage Account	Read attachments
UAMI (api)	SB Data Sender	Service Bus NS	Enqueue messages
UAMI (worker)	Cosmos DB Data Contrib	Cosmos Account	Read/write docs
UAMI (worker)	Storage Blob Reader	Storage Account	Read PDF files
UAMI (worker)	SB Data Receiver	Service Bus NS	Dequeue + complete
UAMI (worker)	Cognitive Svc User	AI Foundry	Call OCR + LLM
UAMI (both)	AcrPull	Container Registry	Pull images
Developers	KV Secrets User	Key Vault	Read secrets

Appendix A: Detailed OCR Performance Metrics

Table 9: Per-Language OCR Accuracy (Test Suite: 500 docs/lang)

Language	Printed	Handwritten	Mixed	Tables	Avg Time
English	99.8%	98.2%	98.9%	99.1%	2.1s
German	99.6%	97.5%	98.4%	98.8%	2.3s
French	99.5%	97.8%	98.5%	98.9%	2.2s
Spanish	99.4%	97.3%	98.2%	98.7%	2.2s
Dutch	99.3%	96.9%	97.8%	98.5%	2.4s
Italian	99.4%	97.1%	98.0%	98.6%	2.3s
Portuguese	99.2%	96.8%	97.7%	98.4%	2.4s
Japanese	98.9%	95.2%	96.8%	97.9%	3.1s
Chinese (Simplified)	98.7%	94.8%	96.5%	97.6%	3.3s
Korean	98.8%	95.0%	96.6%	97.8%	3.0s
Arabic	98.1%	93.5%	95.4%	96.8%	3.5s
Russian	99.0%	96.2%	97.3%	98.1%	2.6s

Table 10: Processing Time Distribution (p50/p90/p95/p99)

Document Type	Pages	p50	p90	p95	p99	Max
Single-page form	1	0.8s	1.2s	1.5s	2.1s	3.2s
Multi-page report	5-10	2.1s	3.4s	4.0s	5.8s	8.1s
Dense tables	1-3	1.5s	2.8s	3.5s	5.2s	7.4s
Scanned (300dpi)	1-5	1.9s	3.1s	3.8s	5.5s	9.0s
Mixed content	10-20	4.2s	6.8s	8.1s	12.3s	18.5s
Large document	30+	8.5s	14.2s	18.0s	25.1s	42.0s
Handwritten	1-3	2.8s	4.5s	5.2s	7.8s	11.2s

Table 11: Token Usage by Model (Monthly Average)

Model	Input Tokens	Output Tokens	Total	Cost	\$/1K tokens
Mistral Doc AI	8,234,500	2,145,300	10,379,800	\$2,076	\$0.20
GPT-4.1-mini	3,456,000	890,500	4,346,500	\$652	\$0.15
GPT-4.1-nano	1,234,000	345,600	1,579,600	\$158	\$0.10
Whisper	-	-	2.3M seconds	\$1,150	\$0.50/hr